

EXP.NO: 9	Develop an Android Calculator Application using Android Studio
DATE: 10.03.2026	

AIM:

To develop an Android application that implements a basic calculator using Android controls such as Button, TextView, and EditText. The calculator will support basic mathematical operations like Addition, Subtraction, Multiplication, and Division.

ALGORITHM:

- 1. Initialize the Android Application:
 - o Set up a new Android project in Android Studio.
 - o Create a MainActivity Java or Kotlin file for the main screen of the application.

- 2. Design the User Interface (UI):
 - o Create the following UI components in the XML layout file:
 - EditText for input fields where the user will enter numbers (two EditTexts for number1 and number2).
 - Buttons for each arithmetic operation (Addition, Subtraction, Multiplication, Division).
 - TextView to display the result of the operation.

- 3. Declare Variables:
 - o Create variables for the EditText fields, Buttons, and TextView.
 - o Define variables to hold the input numbers and the result (e.g., num1, num2, result).

4. Get Input from the User:

- o Get the values entered by the user in EditText fields using `getText().toString()` and convert them to numerical values (like `Integer.parseInt()` or `Double.parseDouble()`).

5. Set Event Listeners for Buttons:

- o For each button (Addition, Subtraction, Multiplication, and Division), set an `OnClickListener`.
- o Inside the `onClick()` method for each button, perform the respective arithmetic operation:

Addition: $\text{result} = \text{num1} + \text{num2}$

Subtraction: $\text{result} = \text{num1} - \text{num2}$

Multiplication: $\text{result} = \text{num1} * \text{num2}$

Division: $\text{result} = \text{num1} / \text{num2}$ (add validation to prevent division by zero).

6. Perform Arithmetic Operation:

- o Check which button was clicked (using `view.getId()`), and based on the button, perform the respective arithmetic operation.
- o Handle Division by Zero: Check if `num2` is 0 before performing division and show an appropriate error message if true.

7. Display the Result:

- o Use `TextView` to display the result of the operation.
- o Convert the result to a string and set it using `setText()` method on the `TextView`.

8. Validate User Inputs:

- o Before performing any operation, validate the input to ensure the user has

entered valid numbers (not empty or non-numeric characters).

9. Handle Edge Cases:

- o Ensure the app handles cases where users input invalid values or attempt division by zero. Display appropriate messages for errors.

10. Finish the Operation:

- o Once the result is displayed, the user can either perform another calculation or exit the app..

CODE:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/num1"
        android:hint="Enter first number"
        android:inputType="numberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/num2"
```

```
android:hint="Enter second number"
android:inputType="numberDecimal"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>
```

<TextView

```
android:id="@+id/result"
android:text="Result"
android:textSize="20sp"
android:padding="10dp"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>
```

<LinearLayout

```
android:orientation="horizontal"
android:layout_width="match_parent"
android:layout_height="wrap_content">
```

```
<Button android:id="@+id/add" android:text="+"
android:layout_weight="1"

    android:layout_width="0dp" android:layout_height="wrap_content"/>
```

```
<Button android:id="@+id/sub" android:text="-"
android:layout_weight="1"

    android:layout_width="0dp" android:layout_height="wrap_content"/>
```

```
<Button android:id="@+id/mul" android:text="*"
android:layout_weight="1"
```

```
        android:layout_width="0dp" android:layout_height="wrap_content"/>

        <Button android:id="@+id/div" android:text="/"
        android:layout_weight="1"

        android:layout_width="0dp" android:layout_height="wrap_content"/>

    </LinearLayout>

</LinearLayout>
```

MainActivity.java:

```
package com.example.calculator;

import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText num1, num2;

    TextView result;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        num1 = findViewById(R.id.num1);

        num2 = findViewById(R.id.num2);
```

```

result = findViewById(R.id.result);

findViewById(R.id.add).setOnClickListener(v -> calculate('+'));
findViewById(R.id.sub).setOnClickListener(v -> calculate('-'));
findViewById(R.id.mul).setOnClickListener(v -> calculate('*'));
findViewById(R.id.div).setOnClickListener(v -> calculate('/'));
}

void calculate(char op) {
    double a = Double.parseDouble(num1.getText().toString());
    double b = Double.parseDouble(num2.getText().toString());
    double res = 0;

    switch (op) {
        case '+': res = a + b; break;
        case '-': res = a - b; break;
        case '*': res = a * b; break;
        case '/':
            if (b != 0) res = a / b;
            else {
                result.setText("Cannot divide by zero");
                return;
            }
    }

    result.setText("Result: " + res);
}
}

```

OUTPUT:



Result:

An calculator is designed using Android studio which uses controls like Button, TextView, EditText having basic functionality like Addition, Subtraction, multiplication, and Division.